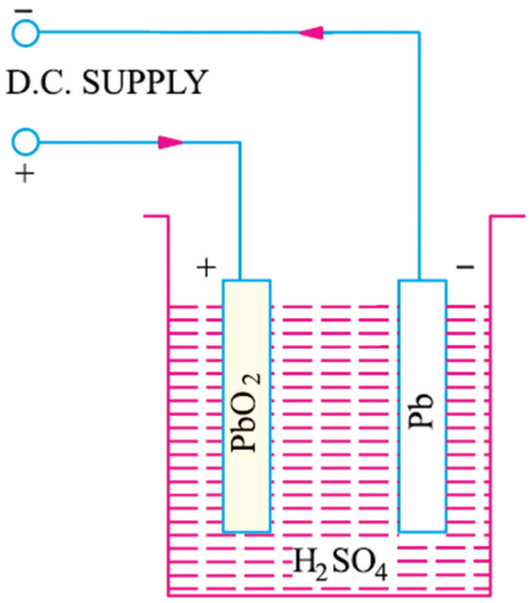



**Fig 4.9** *Charging Process*

### Physical Changes While Charging

1. Anode and cathode return back to their original colour (i.e positive plate dark brown and negative plate grey).
2. Specific gravity of an electrolyte is increased due to absorption of water.
3. Increase in emf values.

### d) Difference Between Primary And Secondary Cells

| Primary Cell | Secondarycell |
|---|---|
| 1. Primary cell cannot be recharged. | Secondary cell can be recharged. |
| 2. Chemical energy is converted into electrical energy. | In this, Electrical energy is converted into chemical energy. |
| 3. Internal resistance is high. | Internal resistance is low. |
| 4. It is light in weight. | It is heavy in weight. |
| 5. It is less expensive. | More expensive. |
| 6. It is disposable. | Not disposable and needs regular maintenance and periodic recharging. |
| 7. Short life. | Long life. |
| 8. Low efficiency. | High efficiency. |
| 9. Less maintenance. | High maintenance. |

## 4.5 LITHIUM - ION BATTERY

A Lithium-ion battery is a type of rechargeable battery in which lithium ions move from the negative electrode to the positive electrode during discharge and lithium ions move from positive electrode to the negative electrode when charging.

The three primary functional components of a lithium ion battery are the positive electrode, negative electrode and electrolyte. The negative electrode is made from carbon. The positive electrode is a metal oxide and electrolyte is a lithium salt in an organic solvent.

Nominal Cell Voltage




**Fig 4.10** *Lithium Ion Battery*

NMC: 3.6/3.85 Volt

Lithium ion battery is a primary cell type battery. (see fig 4.10) It is available in variety of sizes and configurations. Depending on the chemicals used with lithium, the cell voltage is between 2.5 to 3.6volt.



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